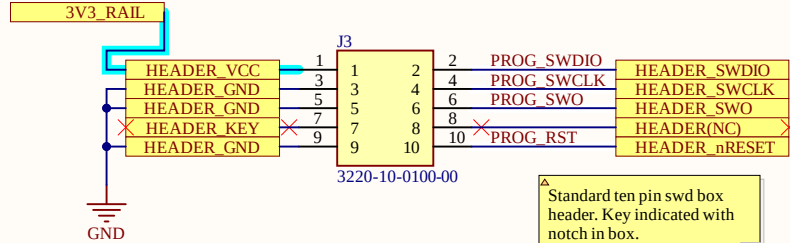
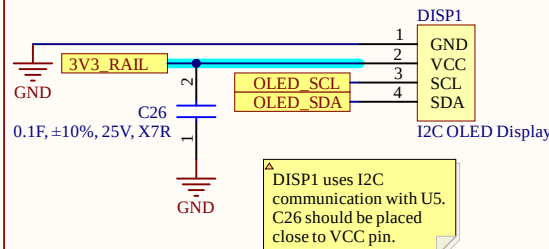


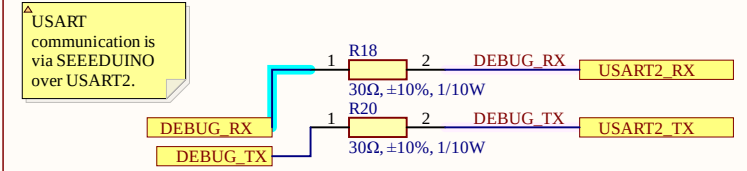
MCU SWD HEADER



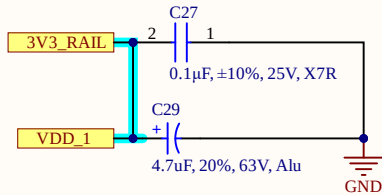
I2C 1.3" OLED DISPLAY



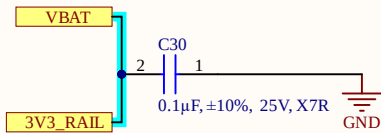
MCU UART TERMINATION RESISTORS



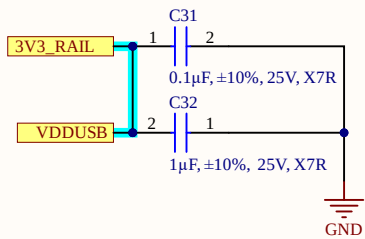
VDD DECOUPLING



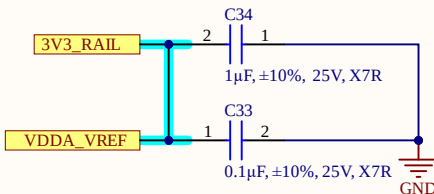
VBAT DECOUPLING



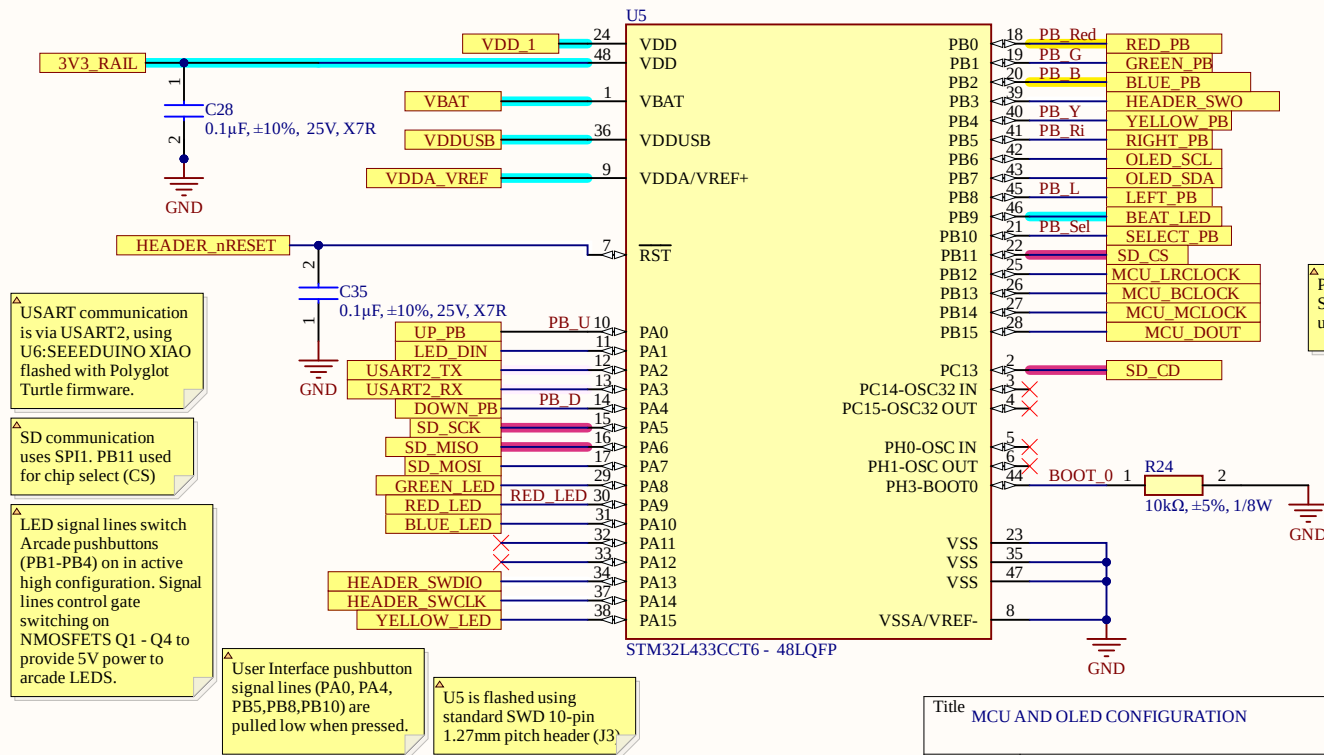
VDD/USB DECOUPLING



VDDA DECOUPLING



STM32L433CCT6 (MCU) PIN CONFIGURATIONS



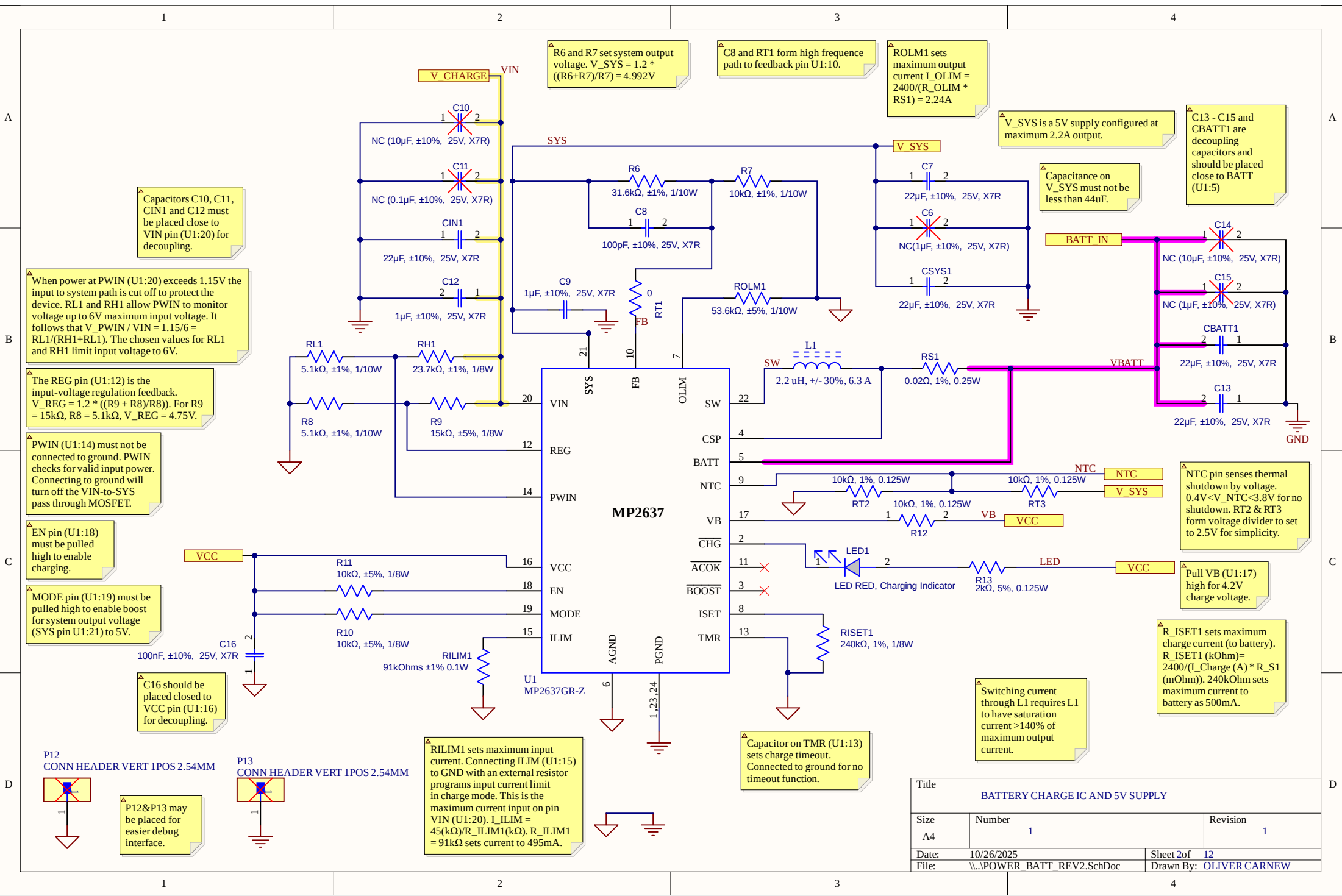
▲ Arcade pushbutton (PB1-PB4) switches pull signal lines low on press. Signal lines trigger unique EXTI vectors, for shorter ISR.

▲ OLED display (DISP1) communication is via I2C, using I2C1.

▲ PB12 - PB15 allows use of SAI1 for audio generation using U3:MAX98357A.

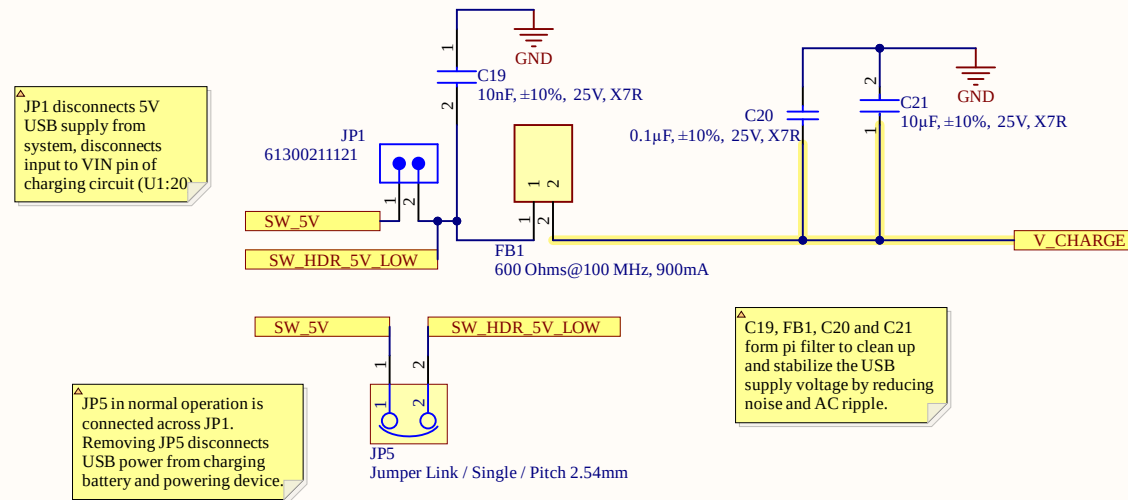
Title: MCU AND OLED CONFIGURATION

Size	Number	Revision
A4	1	1
Date:	10/26/2025	Sheet 1 of 12
File:	\\.\MCU_REV2.SchDoc	Drawn By: OLIVER CARNEW



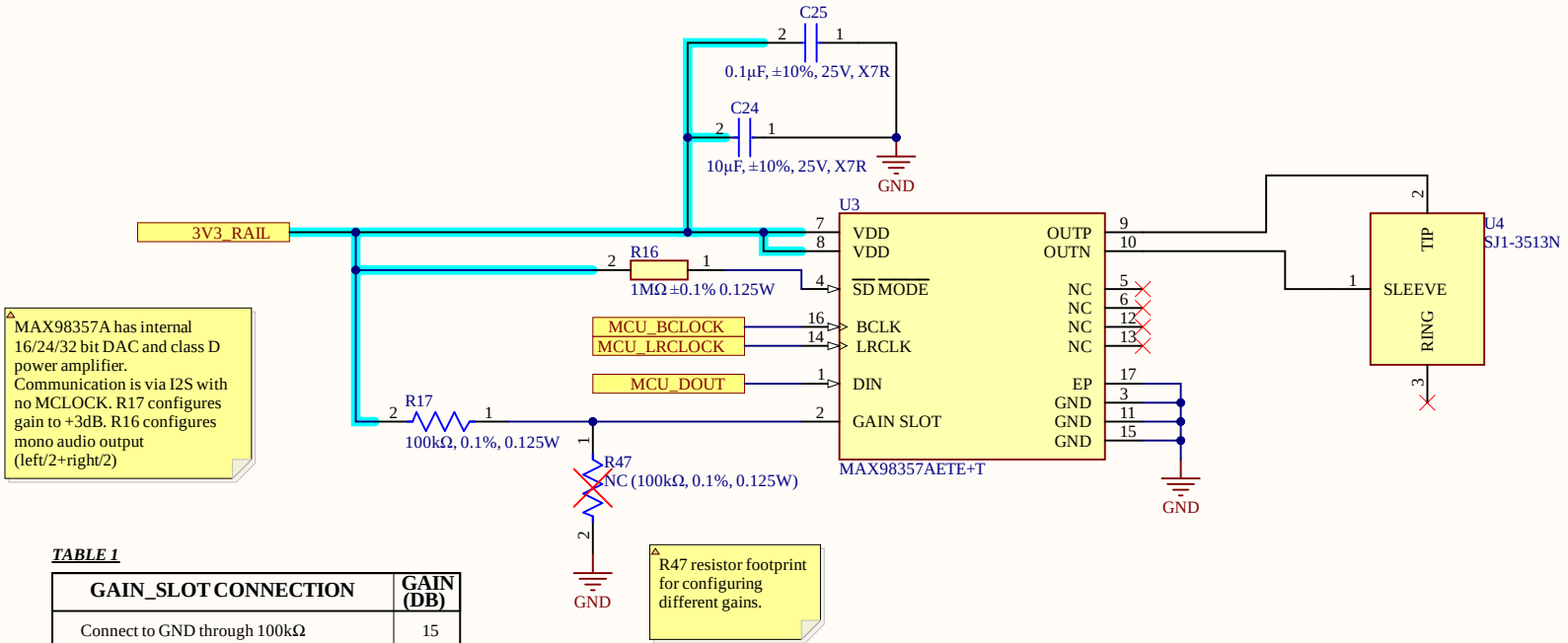
Title		
BATTERY CHARGE IC AND 5V SUPPLY		
Size	Number	Revision
A4	1	1
Date:	10/26/2025	Sheet 2of 12
File:	\\.\POWER_BATT_REV2.SchDoc	Drawn By: OLIVER CARNEW

USB POWER SUPPLY PI FILTER



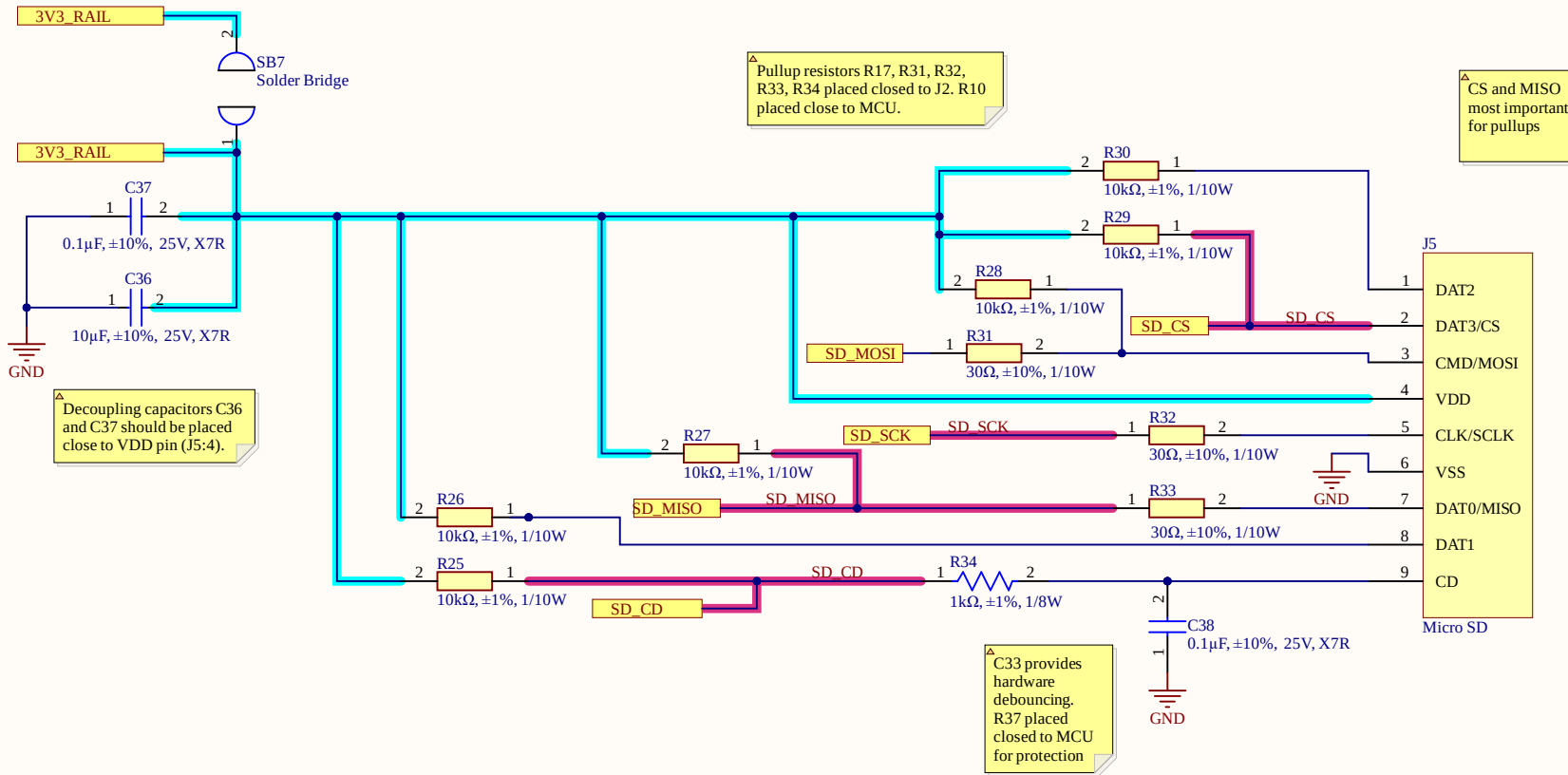
Title		
USB POWER SUPPLY INPUT FILTER		
Size	Number	Revision
A4	1	1
Date:	10/26/2025	Sheet 3 of 12
File:	\\..POWER_REV2.SchDoc	Drawn By: OLIVER CARNEW

AUDIO PLAYBACK DAC AND AMPLIFIER IC CONNECTIONS



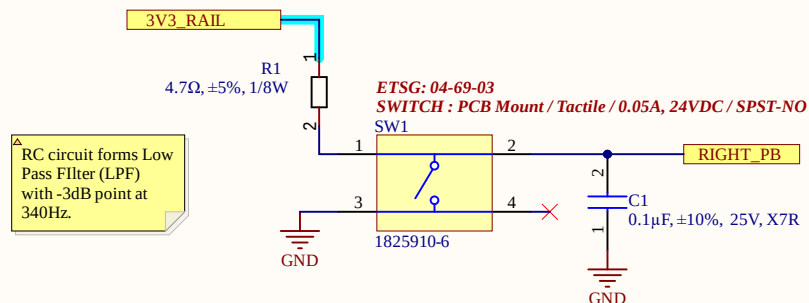
Title		
AUDIO DAC AND AMPLIFIER		
Size	Number	Revision
A4	1	1
Date:	10/26/2025	Sheet 5 of 12
File:	\\.\AUDIO_REV2.SchDoc	Drawn By: OLIVER CARNEW

MICRO-SD CARD SOCKET



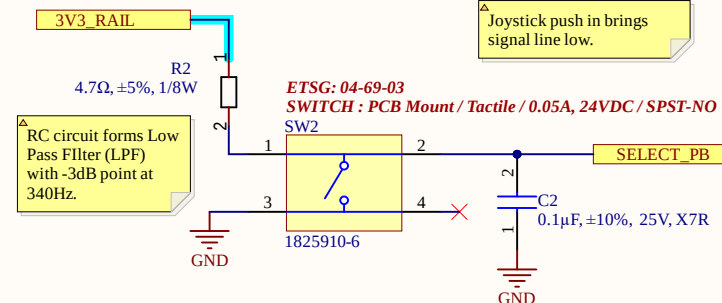
Title		
SD CARD SOCKET		
Size	Number	Revision
A4	1	1
Date:	10/26/2025	Sheet 6 of 12
File:	\\.\SD_CARD_REV2.SchDoc	Drawn By: OLIVER CARNEW

USER INTERFACE MOMENTARY SWITCH RIGHT

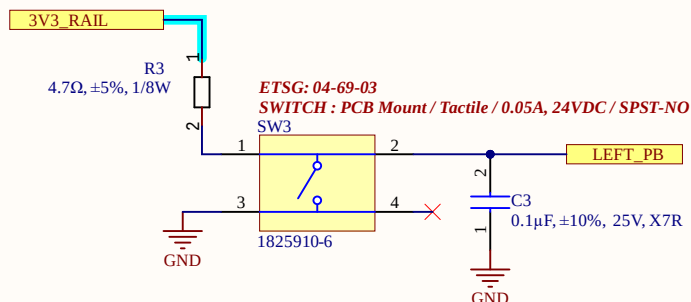


Switches SW1-SW5 are active low, as they pull signal line to ground when pressed.

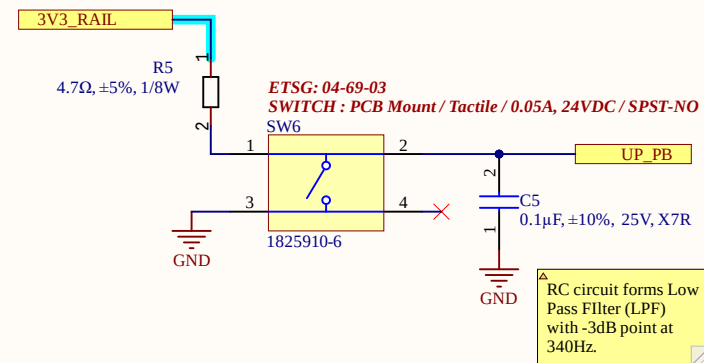
USER INTERFACE MOMENTARY SWITCH SELECT



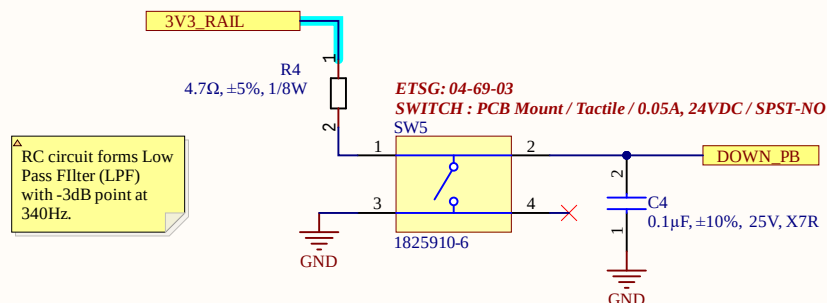
USER INTERFACE MOMENTARY SWITCH LEFT



USER INTERFACE MOMENTARY SWITCH UP

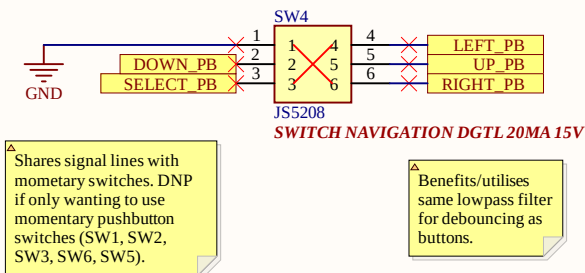


USER INTERFACE MOMENTARY SWITCH DOWN



RC circuit forms Low Pass Filter (LPF) with -3dB point at 340Hz.

USER 5-SWITCH, 2-AXIS JOYSTICK



Title		
USER INTERFACE MOMENTARY SWITCHES		
Size A4	Number 1	Revision 1
Date: 10/26/2025	Sheet 7 of 12	
File: \\.\INTERFACE_rev2.SchDoc	Drawn By: OLIVER CARNEW	

ARCADE PUSHBUTTON LEADS

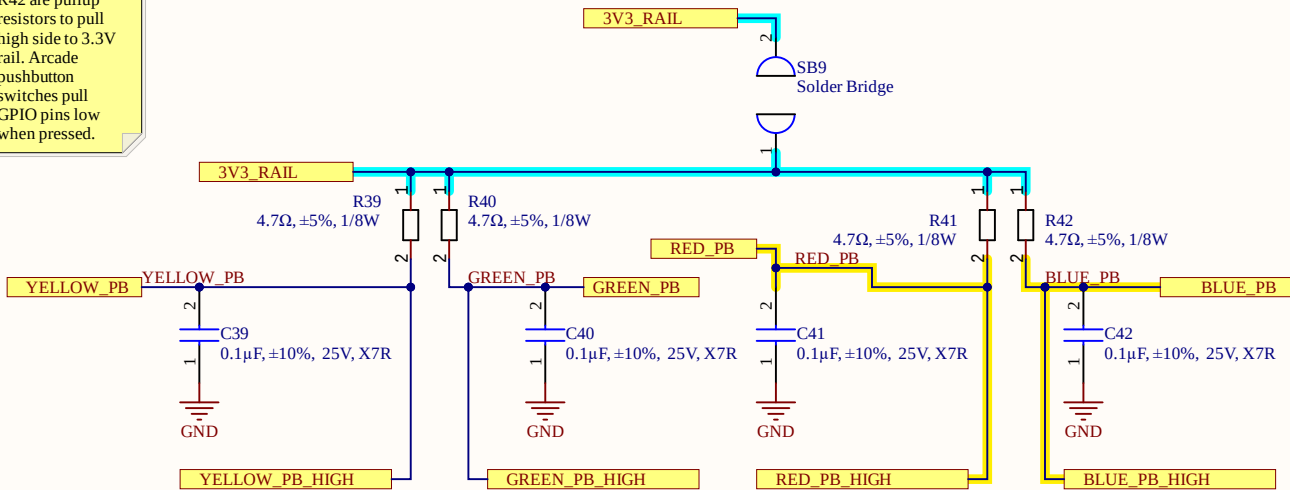


Δ R22, R35, R36, R45 limit gate capacitance current into GPIO pin. 330Ohm chosen to minimise BOM count. Lower values improve switching response time. Higher values increase noise immunity.

All arcade pushbutton leds are powered from 5V system rail. Necessarily, these GPIO controlled leds necessitate voltage controlled MOSFET gate switching (BSS138). LEDs are lowside switched, active high control.

⚠ R37, R43, R44, R46 are gate pulldown resistors. for known state when GPIO gate is high impedance/floating. Lower values provide better immunity but consume more power (voltage divider). 10kOhm reduces BOM count.

ARCADE PUSHBUTTON SWITCHES



p9

4	RED_PB_HIGH
3	GREEN_PB_HIGH
2	BLUE_PB_HIGH
1	YELLOW_PB_HIGH

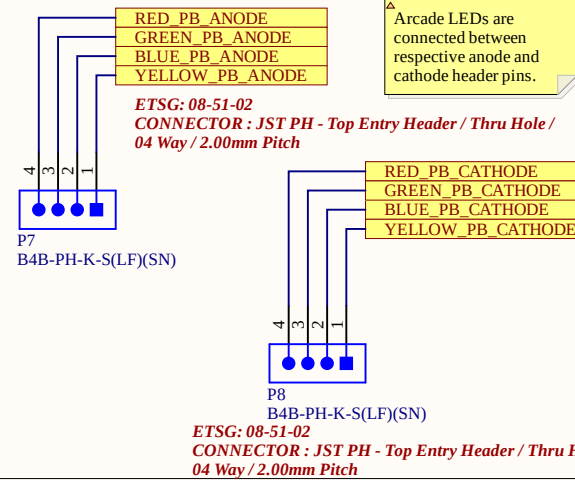
B4B-PH-K-S(LF)(SN)

ETSG: 08-51-02
CONNECTOR : JST PH - Top Entry Header / Thru Hole /
04 Way / 2.00mm Pitch

▲ Arcade pushbutton grounds are all daisy chained together within the wiring harness.

ETSG: 08-51-01 GND
CONNECTOR : JST PH - Top Entry Header / Thru Hole /
02 Way / 2.00mm Pitch

ARCADE LED HEADERS



ARCANE PUSHBUTTON SWITCH AND LED CONTROL		
Size A4	Number 1	Revision 1
Date:	10/26/2025	Sheet 8 of 12
File:	\\.\ARCANE_BUTTONS.SchDoc	Drawn By: OLIVER CARNEW

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FIDUCIALS

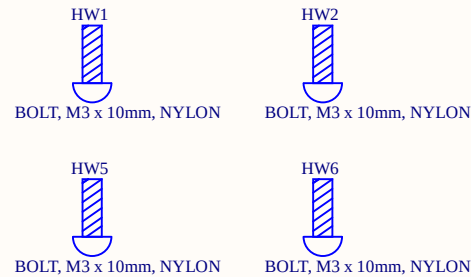
TOP LAYER FIDUCIALS



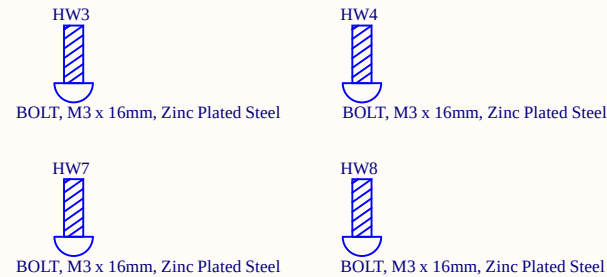
BOTTOM LAYER FIDUCIALS



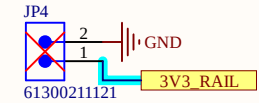
PCB MOUNTING SCREWS



OLED DISPLAY MOUNTING SCREWS



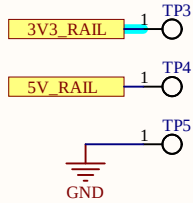
DEBUG



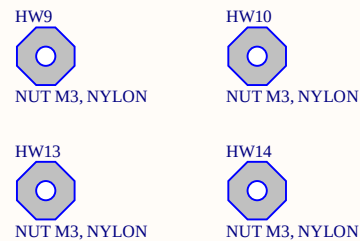
△ DNP for production

△ DEBUG header for easier access to 3.3V rail and GND.

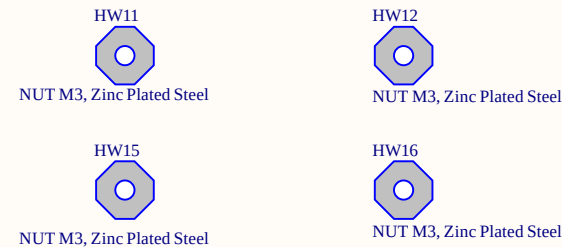
VOLTAGE RAIL TEST POINTS



PCB MOUNTING NUTS



OLED DISPLAY MOUNTING NUTS

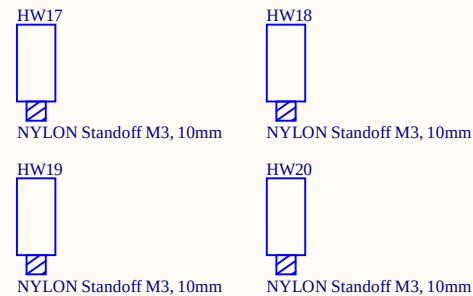


MOUNTING HOLES

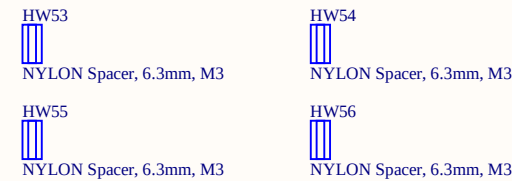


△ All mounting holes should be multiples of 20mm (x-axis&y-axis) from each

PCB MOUNTING STANDOFFS



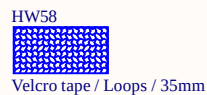
OLED DISPLAY MOUNTING SPACERS



△ HW53-HW56 are threaded onto HW3-HW8 underneath OLED display (DISP1).



ETSG: 28-06-01
ACST : Tape / Velcro / Hook / 35mm / Price per cm



ETSG: 28-06-02
ACST : Tape / Velcro / Loop / 35mm / Price per cm

Title		
MOUNTING HARDWARE AND TESTPOINTS		
Size	Number	Revision
A4	1	1
Date:	10/26/2025	Sheet 9 of 9
File:	\\.\TP_STD_PERIPHERALS.SchDoc	Drawn By: OLIVER CARNEW

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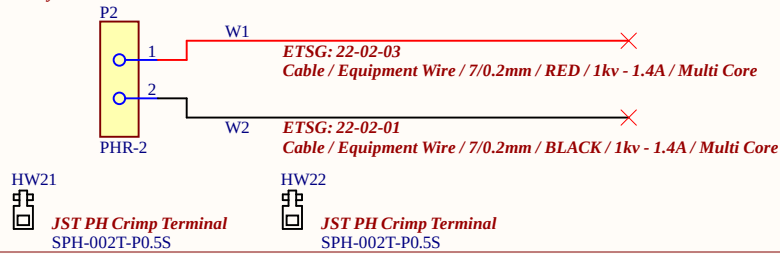
2

3

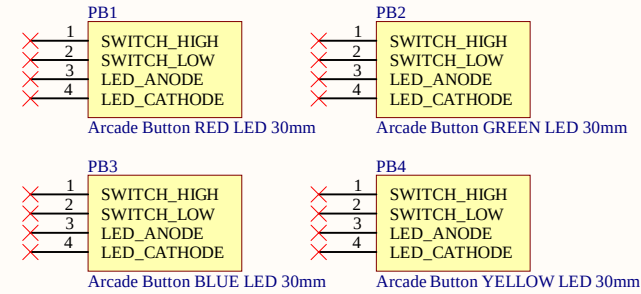
4

ETSG: 08-49-01
JST PH - Crimp Terminal Housing
/ 02 Way / 2.00mm Pitch

BATTERY SIMULATOR CABLE

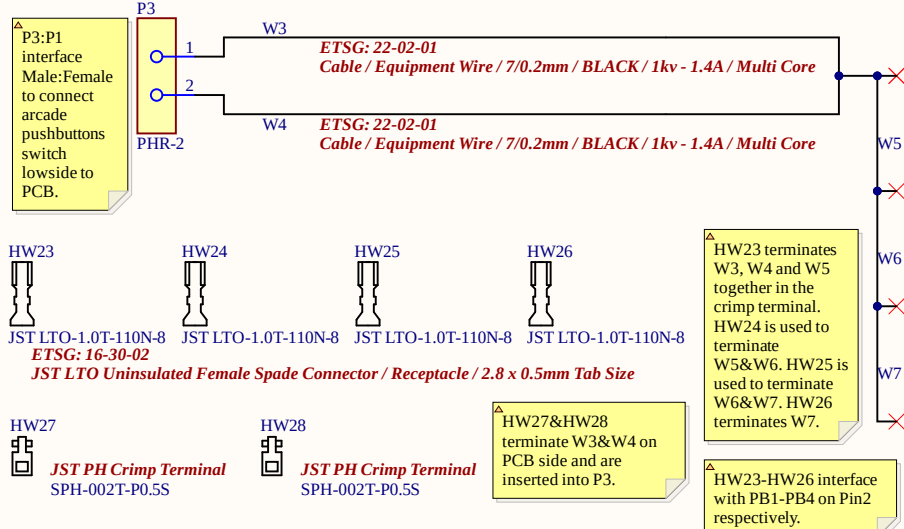


ARCADE PUSHBUTTONS

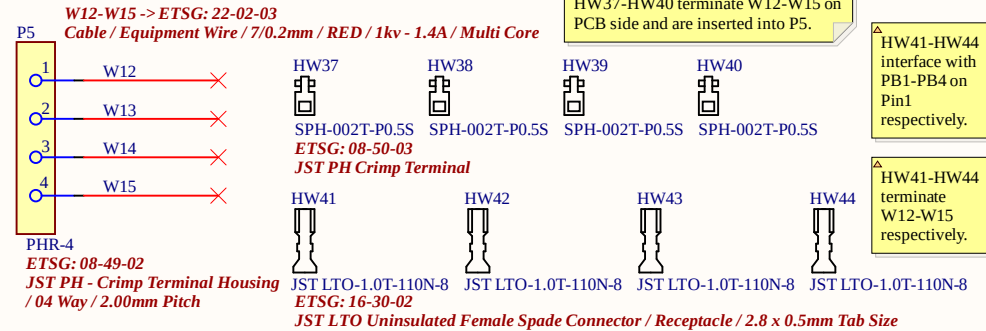


ARCADE PUSHBUTTONS SWITCH LOWSIDE HARNESS

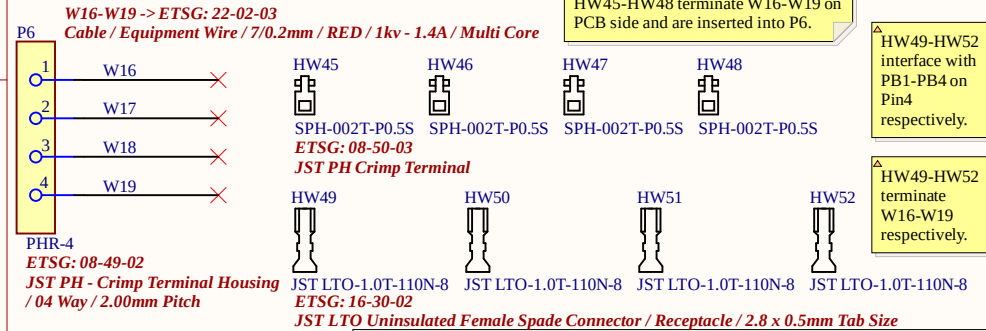
ETSG: 08-49-01
JST PH - Crimp Terminal Housing
/ 02 Way / 2.00mm Pitch



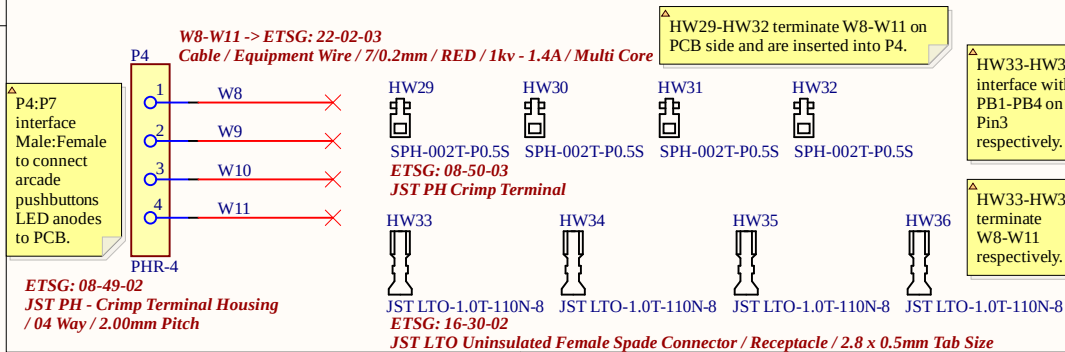
ARCADE PUSHBUTTONS SWITCH HIGHSIDE HARNESS



ARCADE PUSHBUTTONS LED CATHODE HARNESS



ARCADE PUSHBUTTONS LED ANODE HARNESS



Title		
BATTERY AND ARCADE PUSHBUTTON WIRING HARNESSES		
Size	Number	Revision
A4	1	1
Date:	10/26/2025	Sheet 1 of 12
File:	\\.\CONNECTORS_WIRES.SchDoc	Drawn By: OLIVER CARNEW

1

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LED MATRIX CONNECTION CABLE



22-01-3037
ETSG: 01-26-01
CONNECTOR : KK Style - Crimp Terminal Housing / 03 Way / 2.54mm Pitch / White

W20 - W22 are terminated at both ends with KK terminals (HW59-HW64) and inserted into P10 and P11.

22-01-3037
ETSG: 01-26-01
CONNECTOR : KK Style - Crimp Terminal Housing / 03 Way / 2.54mm Pitch / White



ETSG: 01-24-01
CONNECTOR : KK Style - Crimp Terminal / K211



ETSG: 01-24-01
CONNECTOR : KK Style - Crimp Terminal / K211



ETSG: 01-24-01
CONNECTOR : KK Style - Crimp Terminal / K211



ETSG: 01-24-01
CONNECTOR : KK Style - Crimp Terminal / K211

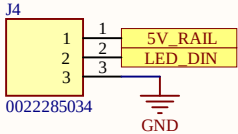


ETSG: 01-24-01
CONNECTOR : KK Style - Crimp Terminal / K211



ETSG: 01-24-01
CONNECTOR : KK Style - Crimp Terminal / K211

LED MATRIX CONNECTION HEADER

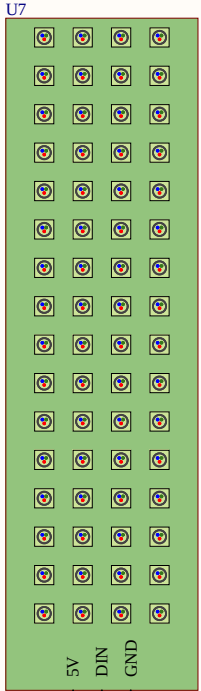


Female 3-Way KK terminal housing P10 interfaces with J4 for LED matrix connection.

Keyed header. 'LED_DIN' is a 3.3V control signal that is shifted to 5V on the LED matrix.

ETSG: 01-34-01
CONNECTOR : KK Style - Straight Header / 03 Way / 2.54mm Pitch / With Friction Lock

LED MATRIX

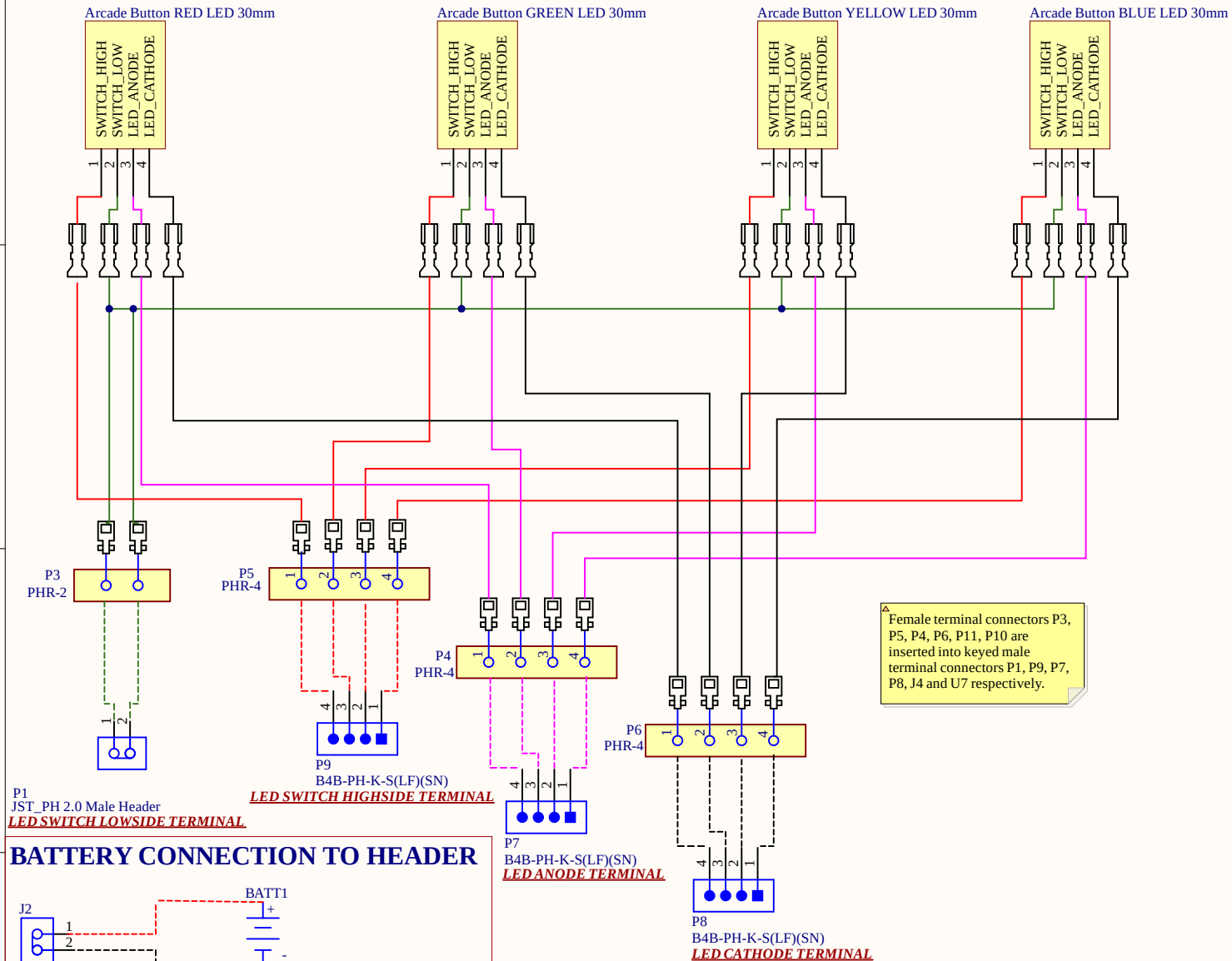


WS2812B LED matrix, 16x4

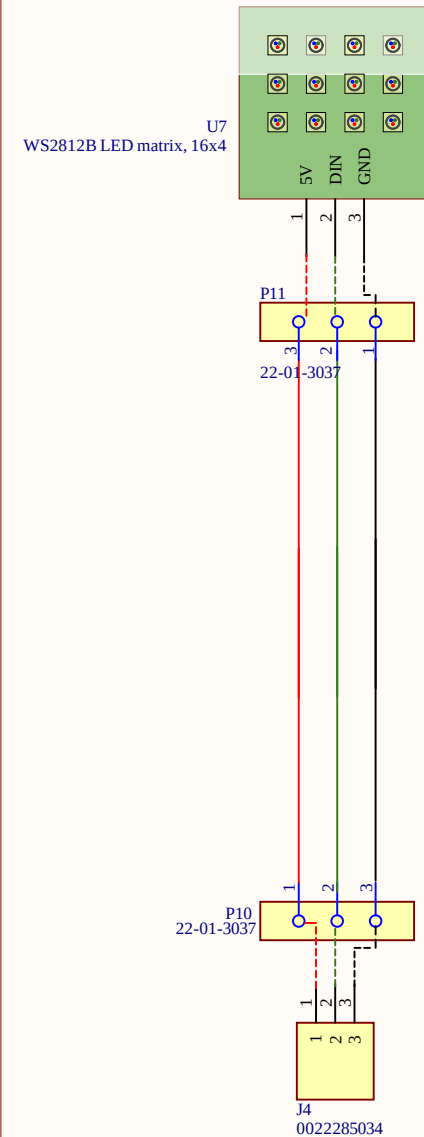
LED matrix on board 3 way KK terminal header interfaces with P11 female connector for connection to PCB board.

Title		
LED MATRIX CONNECTION AND WIRING		
Size	Number	Revision
A4	1	1
Date:	10/26/2025	Sheet 1 of 12
File:	\\.\LED_MATRIX.SchDoc	Drawn By: OLIVER CARNEW

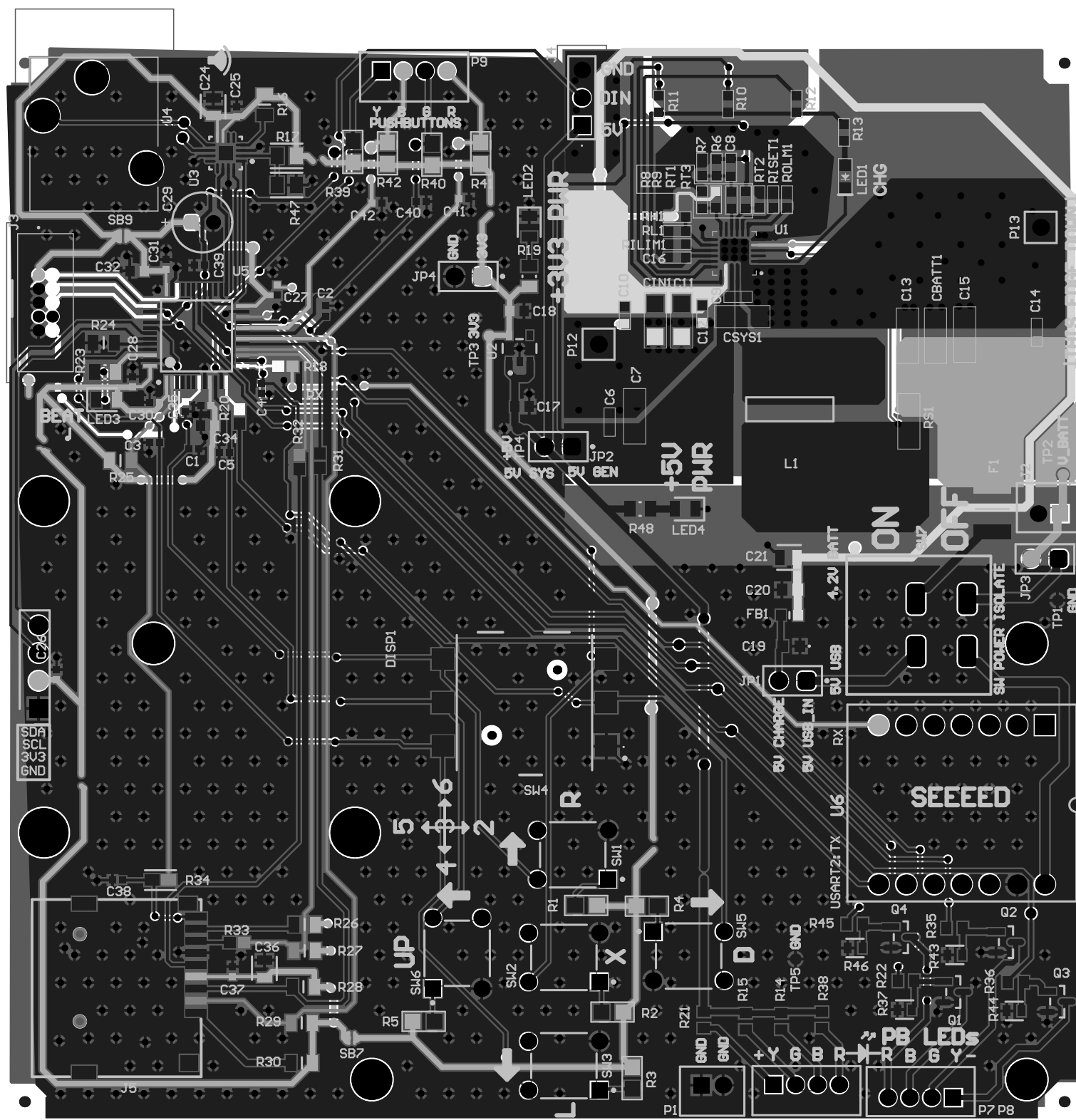
ARCADE PUSHBUTTONS CONNECTION TO HEADERS



LED MATRIX CONNECTION TO HEADER



Title			
BATTERY, ARCADE PUSHBUTTONS AND LED MATRIX WIRING DIAGRAM			
Size	Number	Revision	
A4	1	1	
Date:	10/26/2025	Sheet of	12
File:	\\.\WIRING_DIAGRAM.SchDoc	Drawn By:	OLIVER CARNEW



Board Stack Report